

Network Traffic Threat Hunting Investigation Report – Session 1

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Executive Summary

A 20-minute live network capture was analyzed to detect indicators of compromise, including command-and-control (C2) beaconing, abnormal DNS behavior, unusual data transfer, and suspicious endpoint communication. Traffic behavior and protocol distribution were examined using Conversations, Endpoints, DNS filters, and I/O Graph analysis.

1. Methodology

- Captured ~20 minutes of WiFi traffic.
- Reviewed IPv4 & IPv6 conversations sorted by total bytes.
- Investigated high-volume IPs (10.0.6.126 ↔ 10.0.7.75).
- Filtered DNS traffic (dns) and specific IPs (ip.addr filters).
- Reviewed multicast traffic (ff02::fb).
- Used I/O Graph to evaluate periodic communication.

2. Traffic Analysis & Observations

Internal IPv4 Traffic: High packet exchange observed between 10.0.6.126 and 10.0.7.75 (private range 10.0.0.0/8). Communication consistent with internal network routing. No abnormal port usage detected.

DNS Activity: Queries directed to 8.8.4.4 (Google Public DNS). Domains such as googlevideo.com identified, indicating YouTube CDN traffic. No evidence of DGA-like domains, excessive NXDOMAIN responses, or repetitive suspicious queries.

IPv6 Multicast: Traffic to ff02::fb identified as mDNS (local service discovery). Behavior aligns with normal OS background activity.

I/O Graph Review: Traffic spikes were irregular and user-driven. No fixed periodic intervals (e.g., 30s/60s repetition) observed, reducing likelihood of automated C2 beaconing.

3. Evidence Screenshots

Figure 1

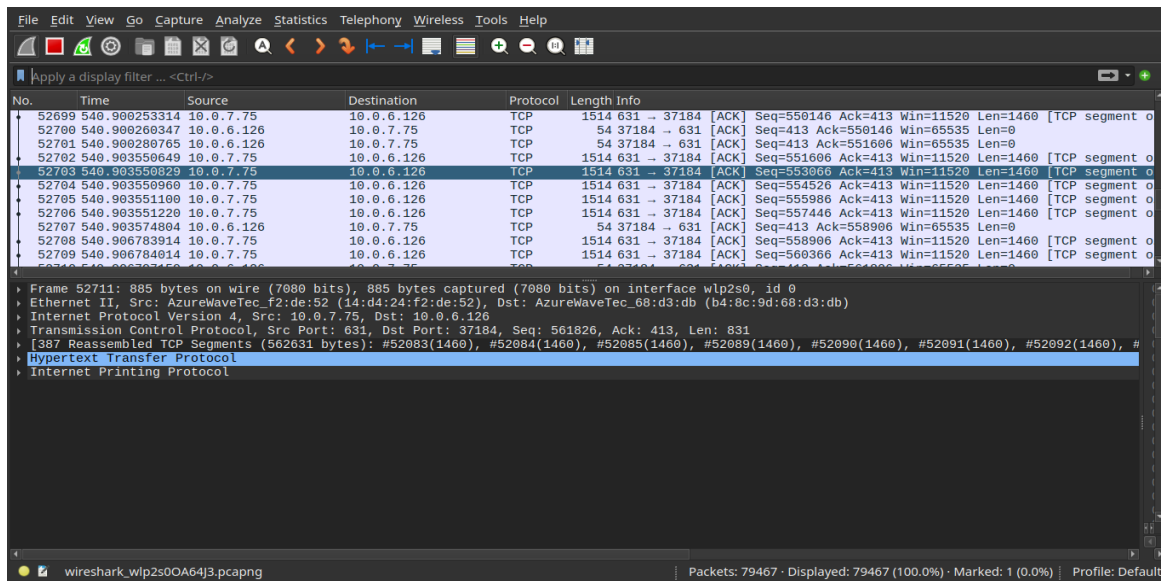


Figure 2

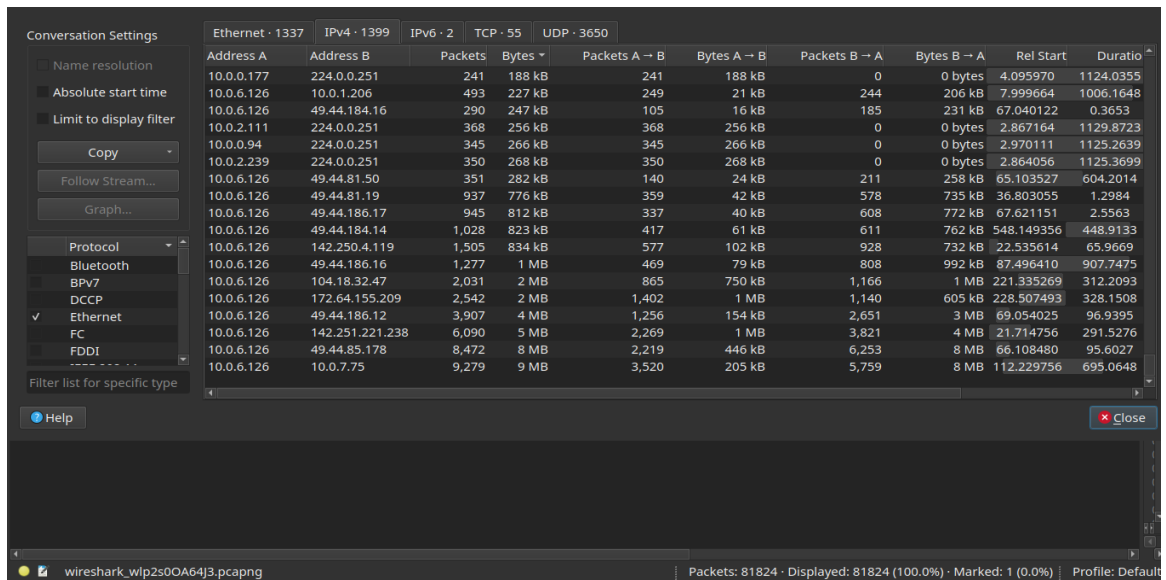


Figure 3

Conversation Settings

☐ Name resolution

☒ Absolute start time

☒ Limit to display filter

Copy

Follow Stream...

Graph...

Protocol

Bluetooth

☐ BPv7

☐ DCCP

☒ Ethernet

☐ FC

☐ FDDI

Filter list for specific type

Help

Ethernet · 1360

IPv4 · 1423

IPv6 · 2

TCP · 57

UDP · 3784

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Du
fe80::5abc:637d:d54f:fcce	ff02::2	1	62 bytes	1	62 bytes	0	0 bytes	171.435591	0.00
fe80::5abc:637d:d54f:fcce	ff02::fb	49	14 kB	49	14 kB	0	0 bytes	23.144868	1151.2

wireshark_wlp2s0OA64j3.pcapng

Packets: 83975 · Displayed: 83975 (100.0%) · Marked: 1 (0.0%) · Profile: Default

Figure 4

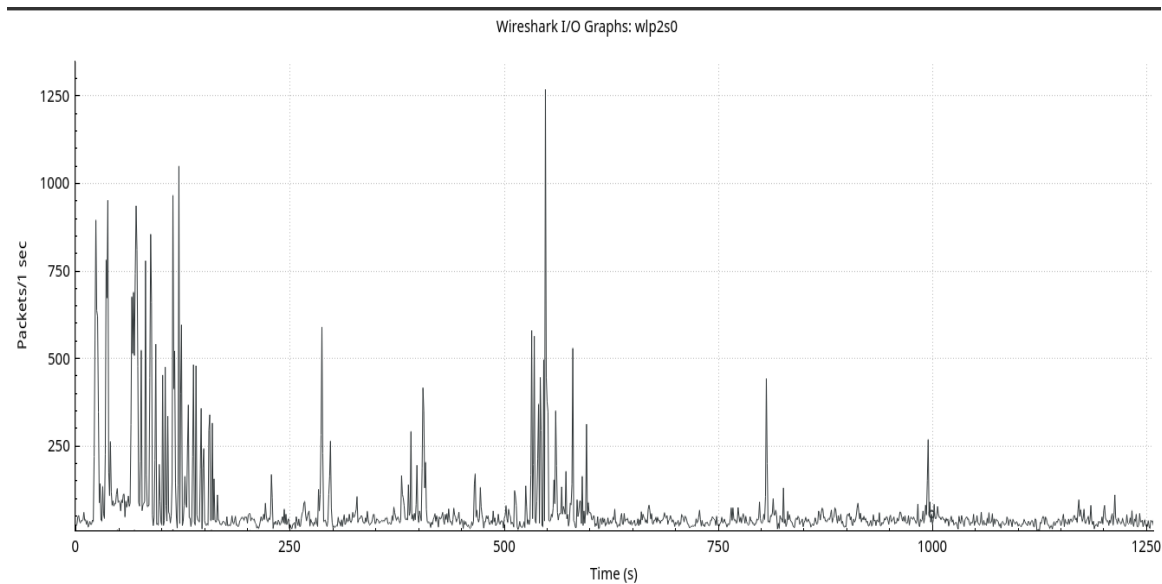


Figure 5

